

Operation Contracts (Updated at the end of Iteration 3)

Contract: Buy Object

Operation: buy(Object object)

Cross References: none for now

Preconditions:

- Object must not already be owned by the institution
- Object must be up for auction

Postconditions:

- The Object is added to the right art area catalog
- Object.Price is instantiated and value is updated
- Object.isOwned is updated to “true”
- Object.isAuctioned is updated to “false”

Contract: Sell Object

Operation: sell(Object object)

Cross References: none for now

Preconditions:

- Object must be up for auction, so Object.isAuctioned should be “true”
- Object must be owned by the institution, so Object.isOwned should be “true”

Postconditions:

- The Object is removed from the catalog
- All values stored concerning object are removed from the database

Contract: add Auction

Operation: addAuction(location: string, Date: String, StartTime: time, EndTime: time,
Speciality: string)

Cross References: none for now

Preconditions:

- No other auction should take place at the same time in the same location

Postconditions:

- A new Auction object is instantiated
- The new auction is added to the catalog
- Auction.location is instantiated and value is updated
- Auction.date is instantiated and value is updated
- Auction.startTime is instantiated and value is updated
- Auction.endTime is instantiated and value is updated
- Auction.Speciality is instantiated and value is updated

Contract: Sign Up

Operation: addClient(username: string, password: string, affiliation: string, info: string,
intent: string)

Cross References: none for now

Preconditions:

- The administrator must approve the information concerning the new client

Postconditions: - - - - -

- A new Client object is instantiated
- The new auction is added to the catalog
- Client.username is instantiated and value is updated
- Client.password is instantiated and value is updated
- Client.affiliation is instantiated and value is updated
- Client.info is instantiated and value is updated
- Client.intent is instantiated and value is updated

Contract: Retrieve information

Operation: retrieved(Object object)

Cross References: none for now

Preconditions:

- Object from which operation wants to retrieve information must be existing
- User must be signed in as either administrator or expert

Postconditions:

- Object's instances and their values are returned

Contract: Offer Availabilities

Operation: addAvailability(Date: String, StartTime: time , EndTime: time)

Cross References: none for now

Preconditions:

- User must be signed in with an expert account
- User must not have any meeting scheduled at the same time
- No auctions in which the user is supposed to attend should be taking place at the same time as that availability

Postconditions:

- A new Availability object is instantiated
- The new availability is added to the catalog
- Availability.date is instantiated and value is updated
- Availability.startTime is instantiated and value is updated
- Availability.endTime is instantiated and value is updated

Contract: Approve Client

Operation: approveClient(client: Client)

Cross References: none for now

Preconditions:

- The client must have an account in "pending" status

Postconditions:

- Client account status is updated to "approved"
- Client is notified of approval (optional)
- Client can now log into the system

Contract: Request Service

Operation: requestService(clientId: string, expertId: string?, serviceType: string, objectId: string?, auctionId: string?)

Cross References: none for now

Preconditions:

- Client must be logged in and approved
- ServiceType must be one of [Consultation, Accompaniment, Representation]

Postconditions:

- A new ServiceRequest object is instantiated
- ServiceRequest is linked to the client (and optionally to an expert)
- ServiceRequest.type is instantiated and value is updated
- Related object or auction references are stored if applicable

Contract: Search Objects

Operation: searchObjects(criteria: string)

Cross References: none for now

Preconditions:

- Client must be logged in

Postconditions:

- A filtered list of ObjectOfInterest entries is returned to the client

Contract: Search Auctions

Operation: searchAuctions(criteria: string)

Cross References: none for now

Preconditions:

- Client must be logged in

Postconditions:

- A filtered list of Auction entries is returned to the client

Contract: Login Client

Operation: loginClient(email: string, password: string)

Cross References: none for now

Preconditions:

- Client account must exist
- Client account must be approved

Postconditions:

- A new session is started for the client
- The client is redirected to their dashboard

Contract: Add Expert

Operation: addExpert(name: string, contact: string, license: string, expertiseAreas: string[])

Cross References: none for now

Preconditions:

- The administrator must be logged in

Postconditions:

- A new Expert object is instantiated
- Expert.name is instantiated and value is updated
- Expert.contact is instantiated and value is updated
- Expert.license is instantiated and value is updated
- Expert.expertiseAreas is instantiated and value is updated
- A system account is created for the Expert

Contract: Update Expert Availability

Operation: updateAvailability(expertId: string, date: string, startTime: time, endTime: time)

Cross References: none for now

Preconditions:

- User must be signed in with an expert or administrator account
- The expert must not have a meeting or availability overlapping with the new slot
- No auctions the expert is scheduled to attend should overlap with the time slot

Postconditions:

- A new Availability object is instantiated
- The new availability is added to the catalog
- Availability.date is instantiated and value is updated
- Availability.startTime is instantiated and value is updated
- Availability.endTime is instantiated and value is updated

Contract: Add Object of Interest

Operation: addObject(description: string, type: string, isOwned: boolean, isAuctioned: boolean)

Cross References: none for now

Preconditions:

- The administrator must be logged in

Postconditions:

- A new ObjectOfInterest is instantiated
- Object.description is instantiated and value is updated
- Object.type is instantiated and value is updated
- Object.isOwned is instantiated and value is updated
- Object.isAuctioned is instantiated and value is updated

Contract: Add Auction House

Operation: addAuctionHouse(name: string, location: string)

Cross References: none for now

Preconditions:

- The administrator must be logged in
- No other auction house exists with the same name and location

Postconditions:

- A new AuctionHouse object is instantiated
- AuctionHouse.name is instantiated and value is updated
- AuctionHouse.location is instantiated and value is updated

Contract: Add Auction

Operation: addAuction(location: string, date: string, startTime: time, endTime: time, specialty: string)

Cross References: none for now

Preconditions:

- No other auction is scheduled at the same location, for the same specialty, and overlapping time

Postconditions:

- A new Auction object is instantiated
- Auction.location is instantiated and value is updated
- Auction.date is instantiated and value is updated
- Auction.startTime is instantiated and value is updated
- Auction.endTime is instantiated and value is updated
- Auction.specialty is instantiated and value is updated